

Module 3, Lesson 1

AI 101 for Mechanical Engineers

Read a Table as a Learning Problem

Lesson Outline

1

**Fundamental
Terminology**

2

**Mathematical
Formulation**

3

**Engineering
Application**

4

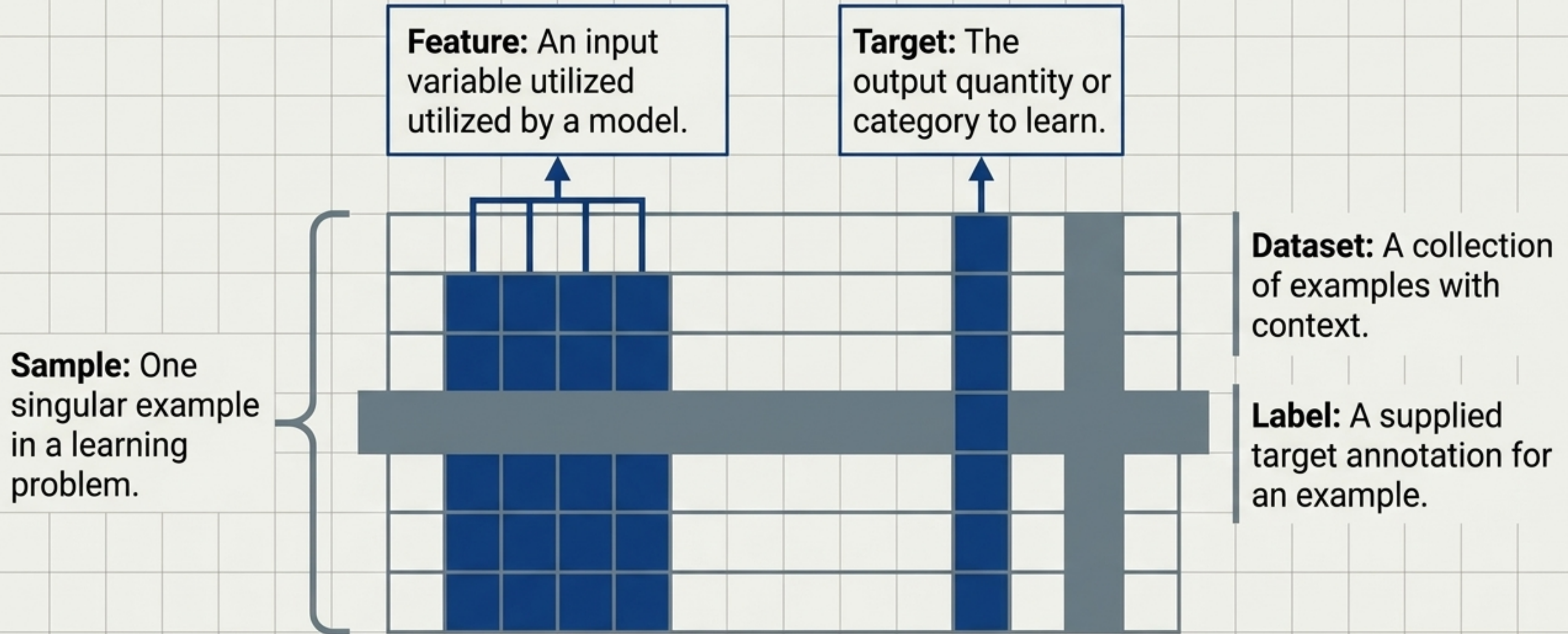
**Concept
Synthesis**

1

1. Fundamental Terminology

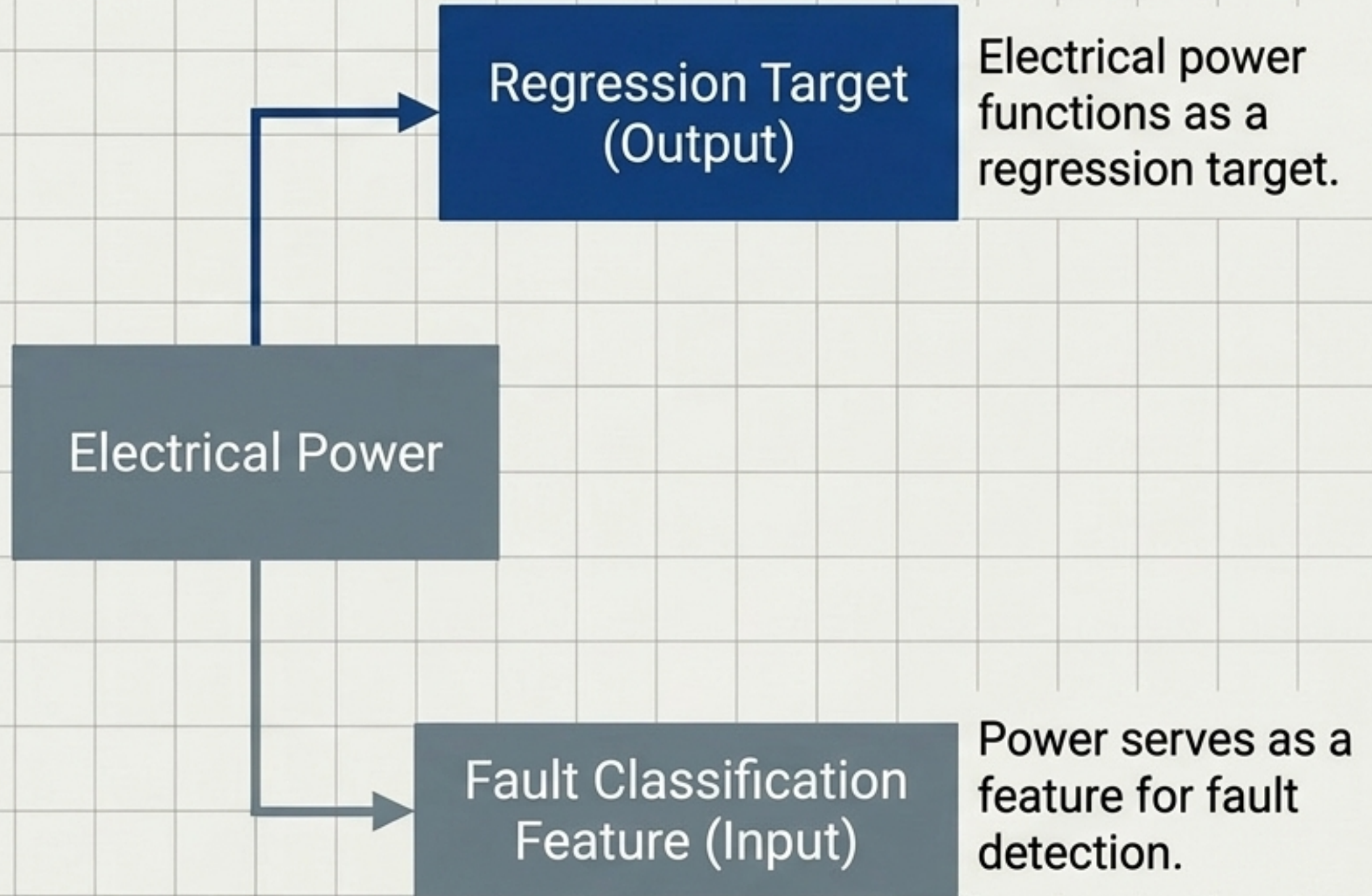
Establish the basic structural units of machine learning datasets.

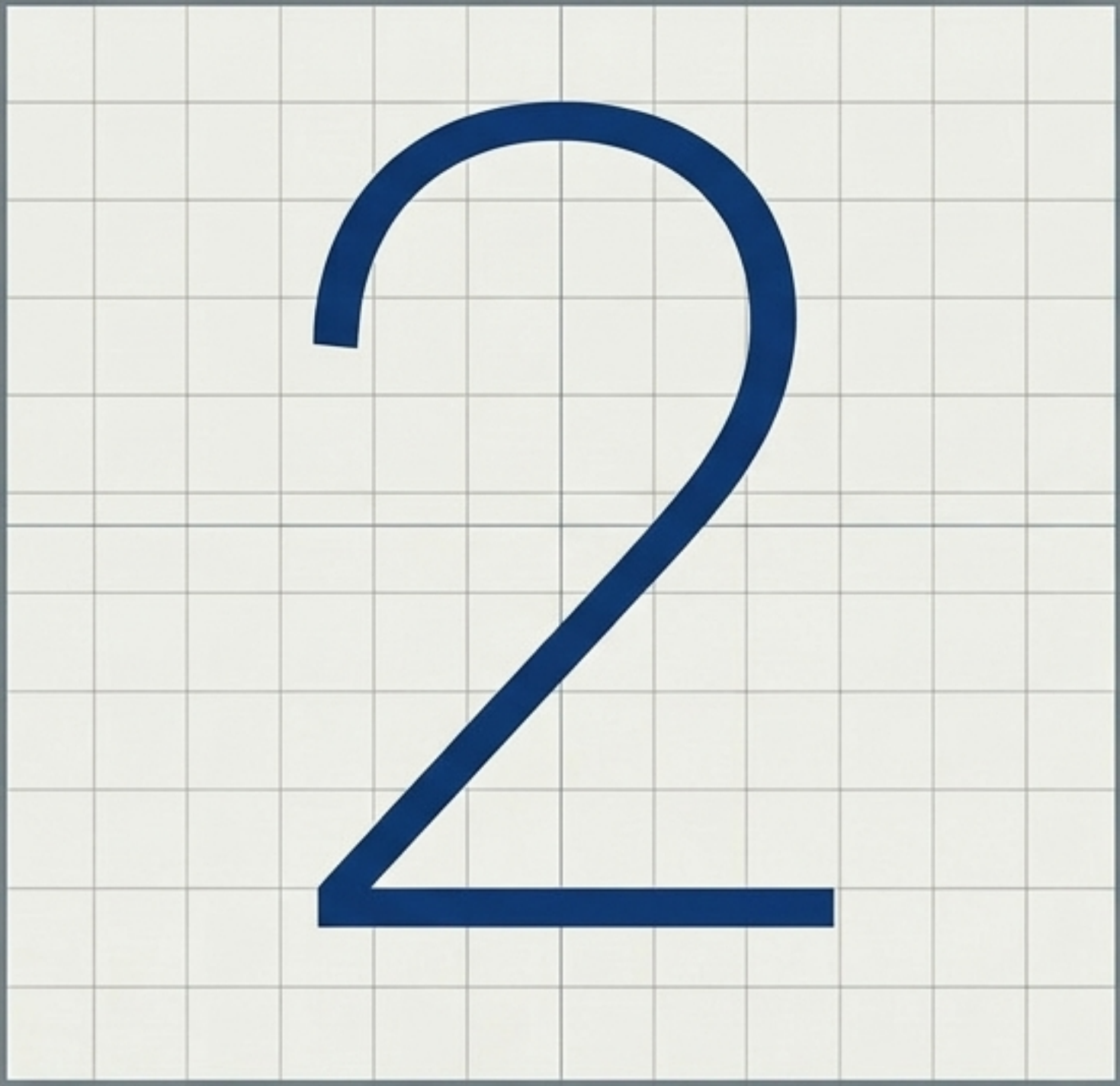
Core Machine Learning Vocabulary



Feature and Target Role Mutability

- Feature roles are not inherently fixed parameters.
- Problem definitions dictate algorithmic data assignment.



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2. Mathematical Formulation

Define exact matrix structures and dimensional shapes required for algorithms.

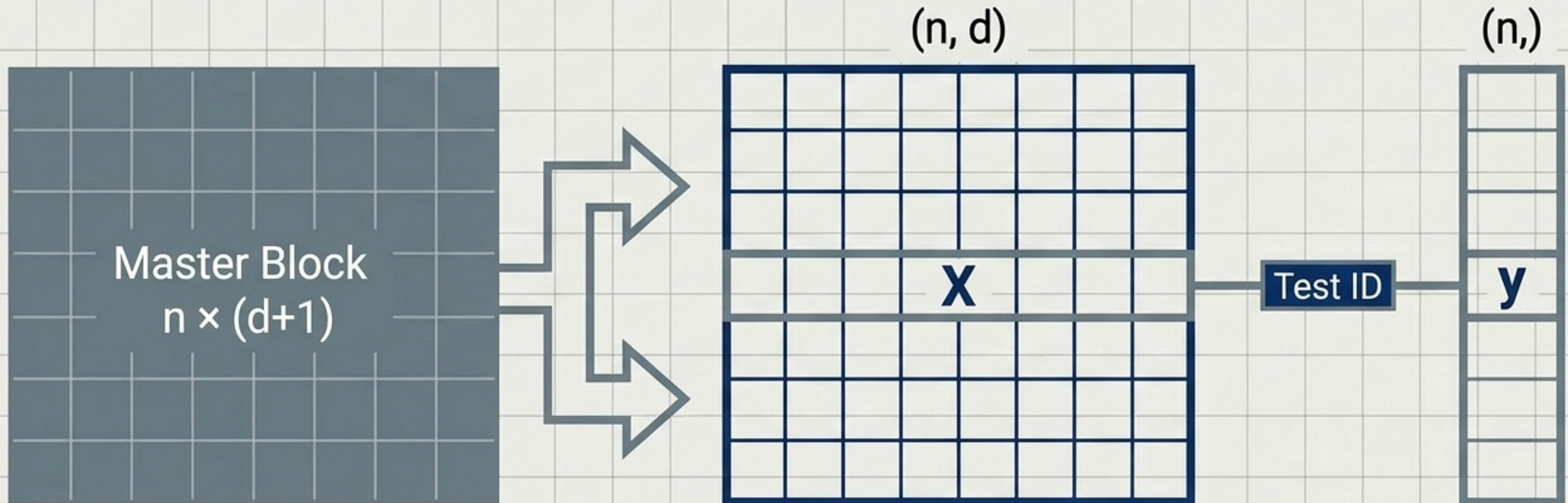
Algorithmic Data Role Matrix

Distinguish matrices to catch accidental data misalignment.
Maintain distinct row and column operational conventions.

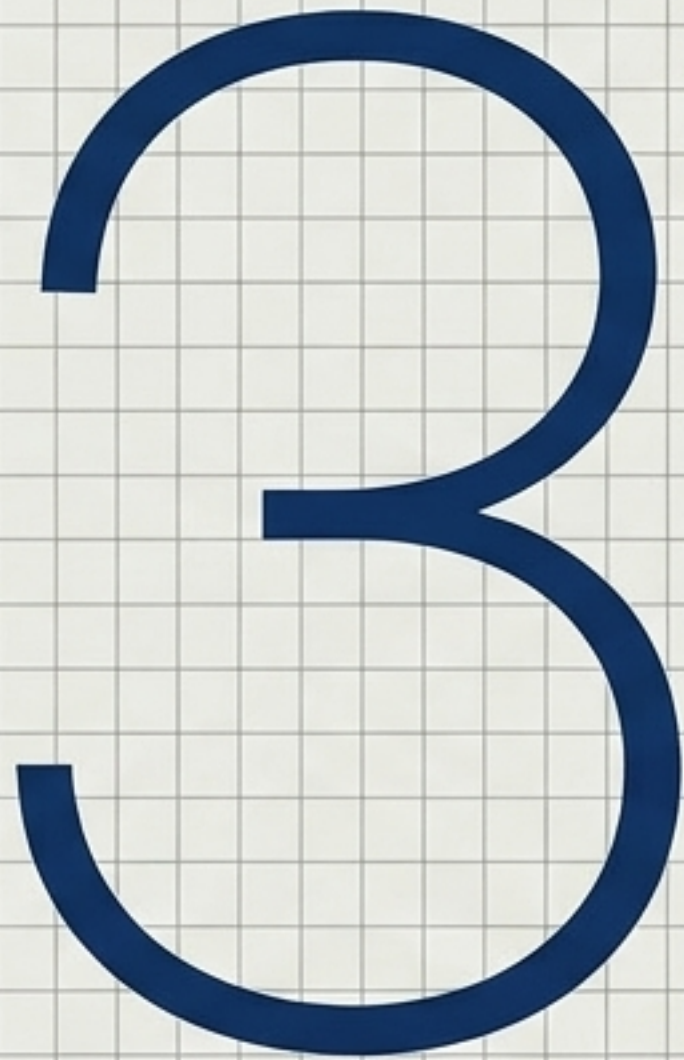
	Feature Matrix (X)	Target Array (y)
Algorithmic Role	Input Variables	Output Quantity
Mathematical Shape	(n, d)	$(n,)$
Physical Example	Pressure Rise [Pa]	Electrical Power [W]

Array Structuring and Dimensionality

- Extract separate analytical arrays from master datasets.
- Matrix **X** contains **n** samples and **d** features.
- Array **y** contains one scalar target per sample.



Array Extraction Flowchart

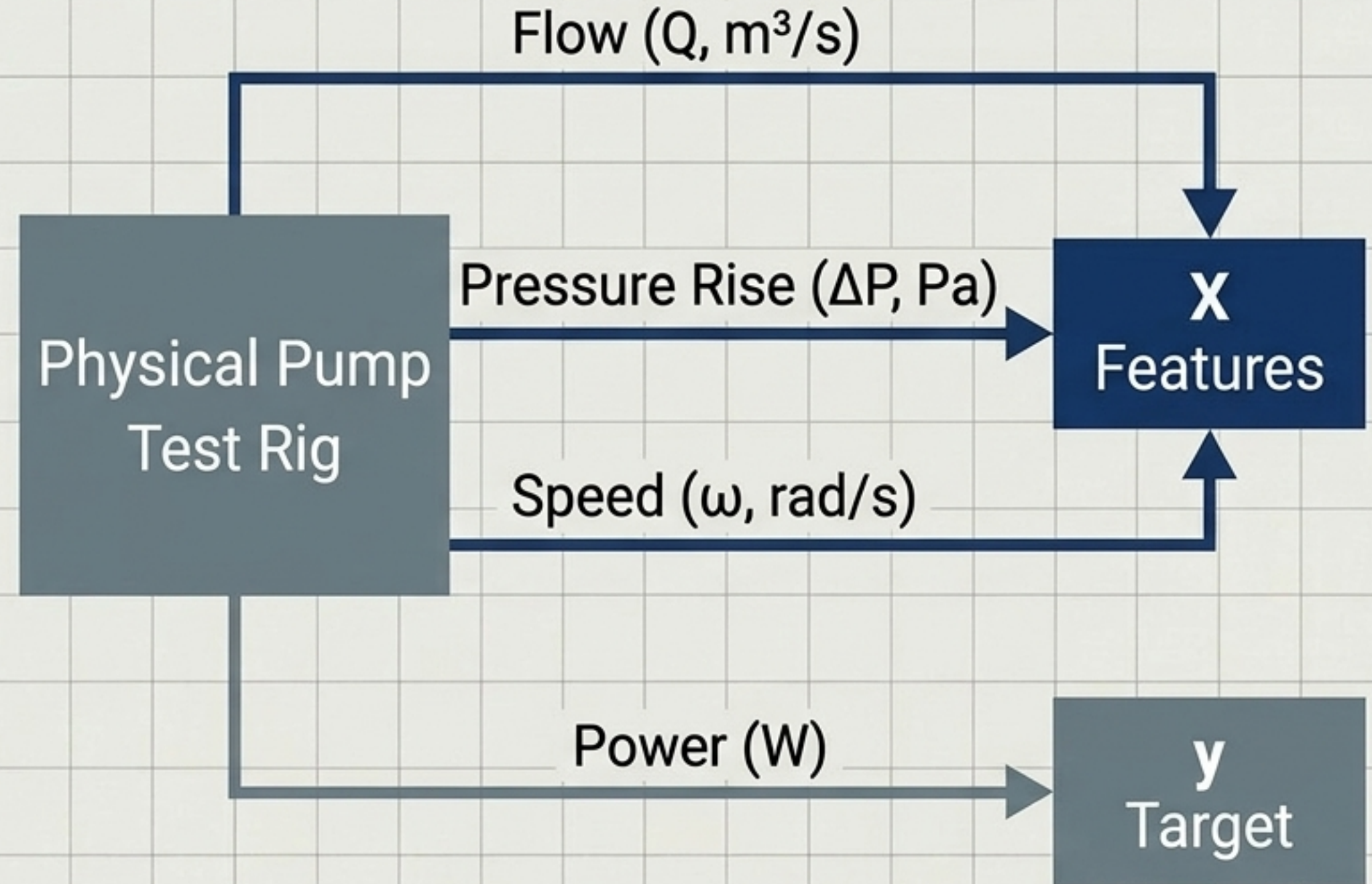
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3. Engineering Application

Map physical pump test data into abstract mathematical structures.

Physical Test Rig Data Mapping

- Analyze 120 independent steady operating pump tests.
- Record fundamental units and calibration metadata precisely.
- Targets represent reference measurements with defined uncertainty.



Matrix Dimensionality Calculation

Matrix shape defines exact sample and feature counts.

$$X_shape = (n, d) \rightarrow (120, 3)$$

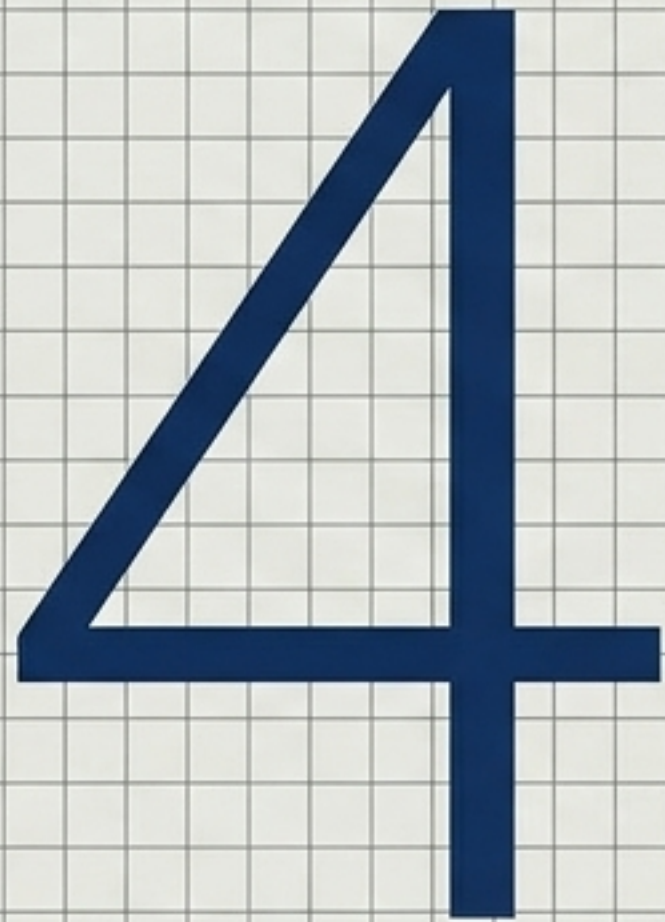
The feature matrix X dimensions are (120, 3).

$$\text{Total Values} = n \times d \rightarrow 120 \times 3 = 360 \text{ Values}$$

The array contains 360 discrete feature values.

$$y_shape = (n,) \rightarrow (120,)$$

The target array y dimensions are (120,).

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4. Concept Synthesis

Review terminology and array constraints holistically.

Structural Concept Reference

- Utilize this reference to verify dataset contracts.
- Ensure metadata tracks acquisition context and conditions.

Array	Shape	Meaning
X	(120, 3)	120 samples with 3 features
y	(120,)	One scalar target per sample
Test ID	120 identifiers	Maintains row alignment

Key Points

Datasets organize physical samples into structural features and targets.

Algebraic shapes separate feature matrices from target arrays.

Physical pump tests map directly to mathematical matrix dimensions.

Test identifiers maintain mandatory row alignment across distinct arrays.